

JEE MAIN REVISION MASTER: HYDROGEN AND IT'S COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Hydrides, Water Hardness & Hydrogen Peroxide	9 PYQs	Inorganic Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hydrides, Water Hardness & Hydrogen Peroxide	$\text{Volume Strength of } \text{H}_2\text{O}_2 = 11.2 \times \text{Molarity} = 5.6 \times \text{Normality}$ $\text{Clark's: } \text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{CaCO}_3 \downarrow + 2\text{H}_2\text{O}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrides, Water Hardness & Hydrogen Peroxide (High Yield — 9 PYQs)

Core Principle: Hydrogen isotopes: Protium (^1H), Deuterium (^2H), Tritium (^3H , radioactive β -emitter). Temporary hardness is caused by bicarbonates of Ca and Mg (removed by boiling or Clark's process with lime). Permanent hardness is caused by chlorides and sulphates of Ca and Mg (removed by washing soda, zeolite or Calgon $\text{Na}_6\text{P}_{10}\text{O}_{18}$). H_2O_2 : non-planar open-book structure; Volume Strength = $11.2 \times M = 5.6 \times N$.

■ **Caution / JEE Pitfall:** Calgon is Sodium hexametaphosphate $\text{Na}_6\text{P}_{10}\text{O}_{18}$; it softens hard water by sequestering Ca^{2+} and Mg^{2+} ions into soluble complex anions.

■ **Speed Trick:** Heavy water D_2O has higher boiling point, density, and viscosity than ordinary water H_2O , and is used as a moderator in nuclear reactors.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following properties is generally not applicable to ionic hydrides?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The correct decreasing order of boiling point of hydrogen halides is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The composition of water gas is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The correct decreasing order of boiling point of hydrogen halides is

- (A) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (B) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (C) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
- (D) $\text{HI} > \text{HF} > \text{HBr} > \text{HCl}$

Correct Answer: None

Solution Analysis: HF is hydrogen bonded, thus it has highest boiling point. The remaining hydrogen halides are gaseous and their boiling points depend on the van der Waals' forces. Greater the molecule mass, greater will be the van der Waals' forces. Hence higher is the boiling point. Thus, the correct order is $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$.